

Complex Solutions to Quadratics

Key Points

To square root negative numbers we use the imaginary number $i = \sqrt{-1}$

A complex number has the form

$$z = x + yi$$

Where x is called the **real** part of z and y is called the **imaginary** part of z . This is wrote sometimes as

$$\text{Re}(z) = x, \text{Im}(z) = y$$

This allows us to solve quadratics which had no real solutions before, when the discriminant is negative

$$\text{i.e } d = b^2 - 4ac < 0$$

Basic Skills

Q1. Convert these negative square roots into imaginary number form

a) $\sqrt{-4}$ b) $\sqrt{-6}$ c) $\sqrt{-8}$ c) $\sqrt{-44}$

Q2. Determine the number of real or complex roots the following quadratics have

a) $x^2 + 4x + 5$

b) $x^2 - 4x - 5$

c) $2x^2 - 3x + 4$

d) $-2x^2 + 10x - 7$

Q3. Solve $x^2 - 8x - 2 = 0$

Competency Skills

Q4. Solve the quadratic equation $x^2 + 9 = 0$

Q5. Solve the quadratic equation $x^2 + 2x + 5 = 0$

Q6. Find the complex roots of the expression $4x^2 - 3x + 5$

Q7. Solve the quadratic equation $3x^2 = x - 2$

Q8. Simplify the following imaginary powers

a) i^2 b) i^3 c) i^4 d) i^5

Advanced Skills

Q9. Simplify the following surds

a) $\sqrt{-6} \times \sqrt{5}$ b) $\sqrt{-20} \times \sqrt{2}$ c) $\sqrt{-8} + \sqrt{-2}$

Q10. The quadratic equation $2z^2 + 6z - 10 = 0$, has solutions z_1, z_2 find $\text{Re}(z_1), \text{Re}(z_2), \text{Im}(z_1), \text{Im}(z_2)$

Q11. The quadratic equation $2z^2 + 6z + 10 = 0$, has solutions z_1, z_2 find $\text{Re}(z_1), \text{Re}(z_2), \text{Im}(z_1), \text{Im}(z_2)$

Q12. Solve the quadratic equation $x^2 + 2x - k = 0$, stating which values of k give a complex number

Q13. Solve the quadratic equation $kx^2 + 2x = k$, stating which values of k give a complex number

Extension

Q14. Can you suggest a formula for i^n for all n

Hint: Use Q7 as a guideline and consider the remainder when n is divided by 4